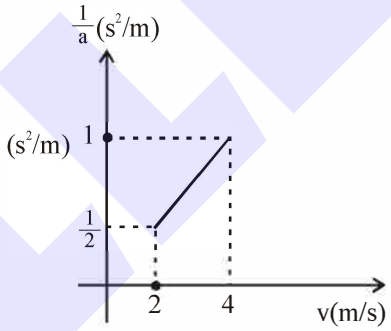
SECTION-I

Single Correct Answer Type **6 Q. [3 M (–1)]**

1. Consider an expanding sphere of instantaneous radius R whose total mass remains constant. The expansion is such that the *instantaneous* density ρ remains uniform throughout the volume. The rate of fractional change in density $\left(\frac{1}{\rho}\frac{d\rho}{dt}\right)$ is constant. The velocity v of any point on the surface of the expanding sphere is proportional to : **[JEE Advanced-2017]**

(A) R^3 (B) $\frac{1}{R}$ (C) R (D) $R^{2/3}$

2. Given graph is $\frac{1}{\text{acceleration}}$ vs velocity graph. If the time interval during which velocity changes from 2m/s to 4m/s is given by Δt seconds. Then find the value of 2Δt :-



(A) 3 (B) 4 (C) 5 (D) 6

3. A boatman moves his boat with a velocity 'v' (relative to water) in river and finds to his surprise that velocity of river 'u' (with respect to ground) is more than 'v'. He has to reach a point directly opposite to the starting point on another bank by travelling minimum possible distance. Then

(A) he must steer the boat (with velocity v) at certain angle with river flow so that he can reach the opposite point on other bank directly.

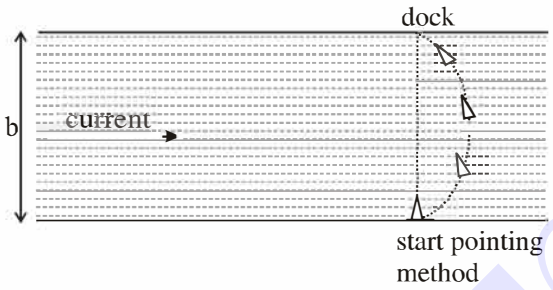
(B) his velocity 'v' must be towards directly opposite point, So, that he can travel rest of distance by walking on other bank to reach the directly opposite point.

(C) boatman should maintain velocity v of boat at certain angle greater than 90° with direction of river flow to minimize drifting and then walk rest of distance on other bank.

(D) boat velocity 'v' should be at an angle less than 90° with direction of river flow to minimize the drift and then walk to the point.

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4. Suppose you are anchored near the shore of a channel in which there is steady current, and you are going to run your (motor) boat at constant throttle to a dock directly across the channel on the opposite shore. The way in which one might steer the boat to the dock is POINTING method. In pointing method keep the nose of the boat pointed directly at the dock. Velocity of boat with respect to water is 'v' and velocity of water is 'u'. Width of water channel is 'b'. The time taken by boat to reach the dock is :-



(A) $\frac{bv}{v^2-u^2}$ (B) $\frac{b}{\sqrt{v^2-u^2}}$ (C) $\frac{bu}{v^2}$ (D) $\frac{bv}{u^2}$

5. When a man started from rest with constant acceleration along an incline plane of angle 37°, from horizontal he found rain to be falling vertically. After t seconds, he finds rain to strike him horizontally. Assume that actual velocity of rain is constant with magnitude 4 m/s. Choose the correct option(s) :
(A) Man is moving up the incline with acceleration of 1 m/s^2 if t = 5 sec.
(B) Man is moving down the incline and if a = 1 m/s^2 then t = 5 sec.
(C) Man is moving down the incline and if t = 20 /3 s then a = 1 m/s^2
(D) Man is moving up the incline and if a = 1 m/s^2 then t = 20/3 s.

Multiple Correct Answer Type **4 Q. [4 M (–1)]**

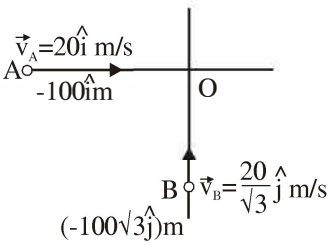
6. In terms of potential difference V, electric current I, permittivity ε_0, permeability μ_0 and speed of light c, the dimensionally correct equation(s) is(are) **[JEE Advanced-2015]**

(A) $\mu_0 I^2 = \varepsilon_0 V^2$ (B) $\varepsilon_0 I = \mu_0 V$ (C) $I = \varepsilon_0 cV$ (D) $\mu_0 cI = \varepsilon_0 V$

7. The motion of a body is given by the equation $\frac{dv(t)}{dt} = 6.0 - 3\ v(t)$; where v (t) is the speed in m/s & t in sec., if the body was at rest at t = 0 : **[JEE '95, 2]**

(A) the terminal speed is 2.0 m/s
(B) the magnitude of the initial acceleration is 6.0 m/s²
(C) the speed varies with time as $v(t) = 2(1 - e^{-3t})$ m/s
(D) the speed is 1.0 m/s when the acceleration is half the initial value .

8. Positions of two vehicles A and B with reference to origin O and their velocities are as shown.

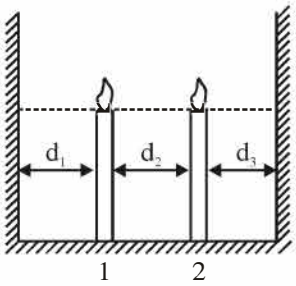


(A) they will collide.
(B) distance of closest approach is 100 m.
(C) their relative speed is $\frac{40}{\sqrt{3}}$ m/s
(D) their relative velocity is $\frac{20}{\sqrt{3}}$ m/s

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9. While making this paper, I stayed up till late at night working with candle light. I had two candles of equal length L lit and placed at the same time, as shown in the figure. Immediately thereafter I noticed that the shadow of the first candle on the left wall is not changing in length, and the shadow of the second candle on the right wall is being shortened at a speed v.

(Take d_1 = 10 cm, d_2 = 20 cm, d_3 = 30 cm, v = 1 mm/s, L = 20 cm.)



(A) The left candle will be burning for longer time than right candle.

(B) The candle to first extinguish will be extinguished at time t = 400 sec.

(C) The candle to extinguish second will be extinguished at time t = 800 sec.

(D) The right candle is decreasing in length at a faster rate.

10. River is flowing with a velocity $\vec{v}_R = 4\hat{i}\ ms^{-1}$. A boat is moving with a velocity $\left(-2\hat{i}+4\hat{j}\right) ms^{-1}$ relative to river and width of river is 100 m along y direction then-

(A) The time of crossing is 25 sec. (B) Absolute velocity of boat is $2\sqrt{5}\ ms^{-1}$

(C) Drift of boat is 50 m. (D) The boat can never cross the river.

Linked Comprehension Type **(2 Para × 3Q.) (1 Para × 2Q.)[3 M (-1)]**

(Single Correct Answer Type)

Paragraph for Question no. 11 to 13

In a certain system of absolute units the acceleration produced by gravity in a body falling freely is denoted by 5, the kinetic energy of a 500 kg shot moving with velocity 400 metres per second is denoted by 2000 & its momentum by 100.

11. The unit of length is :-

(A) 15 m (B) 50 m (C) 25 m (D) 100 m

12. The unit of time is :-

(A) 10 s (B) 20 s (C) 5 s (D) 15 s

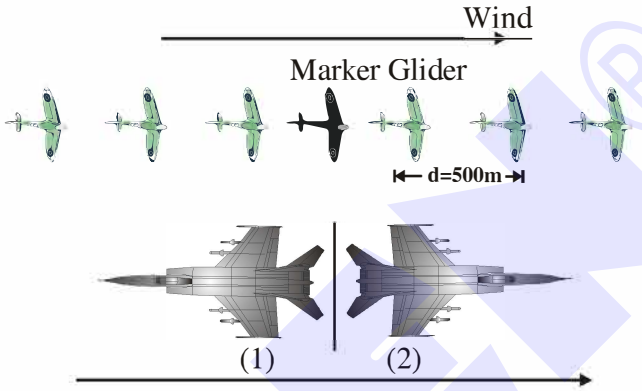
13. The unit of mass is :-

(A) 200 kg (B) 400 kg (C) 800 kg (D) 1200 kg

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Paragraph for Question no. 14 to 16

In an air show a unique exercise was conducted. A large number of gliders [without engine] were released in a strong horizontal wind that imparts a constant velocity of 5 m/s to the gliders. Gliders are travelling in a straight line with a constant separation between each glider. One of the gliders is painted differently and is referred as marker glider. At t = 0, two airplanes were set to move in opposite direactions, one along the wind and other opposite to the wind. Airplane engines maintain a constant speed of 10 m/s relative to the wind. Planes started from marker glider. After 10 minutes of flying, both the airplanes were signalled by marker glider to return and meet at its position. Ignore the length of airplane and gliders in calculation.



14. Mark the incorrect statement.
(A) Both the planes reach the marker glider simultaneously.
(B) Airplane (1) reaches earlier than other plane
(C) For an observer standing on ground, distance traveled by both the airplanes is same
(D) For an observer on marker glider, distance traveled by each airplanes is same

15. Time takes by airplane (1) to return to marker glider after getting signal.
(A) 10 minutes (B) 15 minutes (C) 20 minutes (D) 25 minutes

16. Mark the correct statement
(A) Number of gliders crossed by airplane (1) is 24
(B) Airplane (1) has crossed more gliders than (2) in entire journey.
(C) Airplane (2) has crossed more gliders than (1) in entire journey.
(D) Number of gliders crossed by airplane 2 is 48

Paragraph for Questions 17 and 18

At t = 0 on the planet "Gravitus Increasicus", a projectile is fired with speed 6 m/s at an angle 30° above the horizontal. This planet is a strange one, in that the acceleration due to gravity increases linearly with time, starting with a value of zero when the projectile is fired. In other words, g(t) = 2t.

17. The time of flight of the projectile is :-
(A) 1 sec (B) 2 sec (C) 3 sec (D) $3\sqrt{3}$ sec

18. Range of the projectile is :-
(A) 9m (B) $\frac{9}{\sqrt{3}}$ m (C) $6\sqrt{3}$ m (D) $9\sqrt{3}$ m

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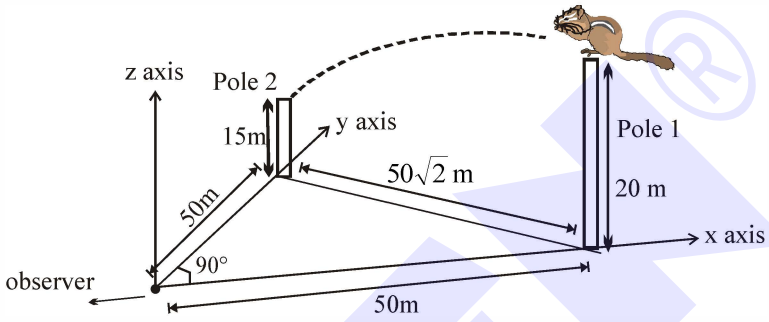
SECTION-II

Numerical Answer Type Question **3 Q.[3(0)]**

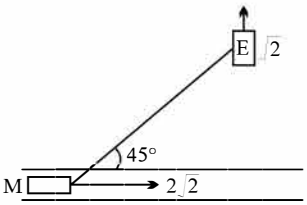
(upto second decimal place)

19. A bird is at a point P (4, –1, –5) and sees two points P_1 (–1, –1, 0) and P_2 (3, –1, –3). At time t = 0, it starts flying with a constant speed of 10 m/s to be in line with points P_1 and P_2 in minimum possible time t. Find t (in sec), if all coordinates are in kilometers.

20. A small squirrel jumps from pole 1 to pole 2 in horizontal direction. Squirrels is observed by a very small observer at origin. What is average velocity vector of squirrel ? If average velocity vector is expressed as $v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$, express your answer as sum of magnitudes of its components $|v_x| + |v_y| + |v_z|$ in unit m/s.



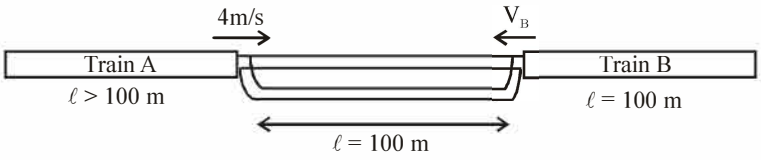
21. A bow man is riding on a horse moving with speed of $2\sqrt{2}$ ms^{-1} along a straight road. He aims at his enemy moving perpendicularly to the road at speed of $\sqrt{2}$ ms^{-1}. At the instant when he fires the arrow, the line joining man and his enemy makes an angle of 45° with the road. Find the angle (in degree) with the road at which he should aim to hit his enemy? Muzzle velocity of arrow is 5ms^{-1}. (given that sin 37° = 3/5). Neglect effect of gravity.



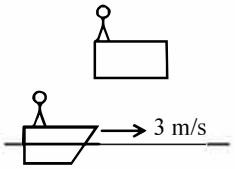
SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **4 Q. [4 M (0)]**

22. Two trains A and B are moving on a straight track towards each other. At t = 0, the situation is as shown in the diagram below. The driver of train B realises that in order to avoid a collision, B must fit into the side track. Length of side track is equal to the length of the train B and length of train A is large as compared to side track. The train A continues with the same speed of 4 m/s. The magnitude of minimum decleration of train B in order to avoid the collision is $\frac{\alpha}{25}$ m/s^2. Then α is.

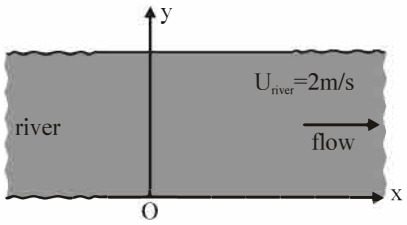


23. You are standing on the chambal Bridge watching the boats in the river. You see a motorboat pass directly below you, traveling perpendicular to the bridge at a speed of 3 m/s. A person on the boat throws a baseball at an initial speed of v_0 and at an angle of 37° from the vertical (Note: both v_0 and the angle are with respect to the boat). Find the value of v_0 (in m/s) necessary for the ball to travel straight up towards you.

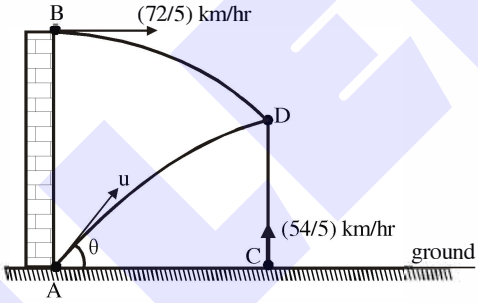


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24. A man can swim in still water with a speed of 3 m/s. x and y axis with respect to ground are drawn along and normal to the bank of river flowing to right with a speed of 2 m/s. The man starts swimming from origin O at t = 0 second. Assume size of man to be negligible. Locus of all the possible points where man can reach at t = 2 sec is $(x - a)^2 + y^2 = c^2$. Find value of $\frac{ac}{4}$.



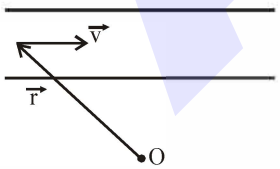
25. In the given figure points A and C are on the horizontal ground & A and B are in same vertical plane. Simultaneously bullets are fired from A, B and C and they collide at D. The bullet at B is fired horizontally with speed of $\frac{72}{5}$ km/hr and the bullet at C is projected vertically upward at velocity of $\frac{54}{5}$ km/hr. Find velocity of the bullet projected from A in m/s.



SECTION-IV

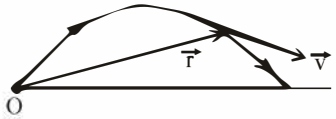
Matrix Match Type (4 × 5) **2 Q. [8 M (for each entry +2(0)]**

26. Column-I shows motion of a particle ; where $\vec{r}$ = position vector of particle, $\vec{v}$ = velocity vector of particle and O represents origin.

(A) 

A car moving along a straight road with constant speed

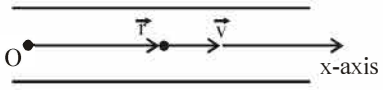
(P) $\frac{d\vec{r}}{dt} \neq 0$

(B) 

A projectile in gravitational field

(Q) $\frac{d\vec{v}}{dt} \neq 0$

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(C) 

A car moving along a straight road with constant speed

(R) $\frac{d|\vec{v}|}{dt} \neq 0$

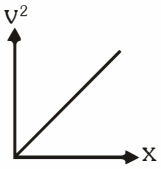
(D) 

A ball projected vertically up under gravity

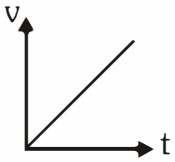
(S) $\left|\frac{d\vec{v}}{dt}\right| = 0$

(T) $\frac{d|\vec{r}|}{dt} \neq 0$

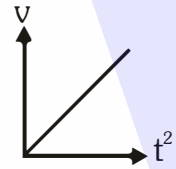
27. **Column-I** **Column-II**

(A) 

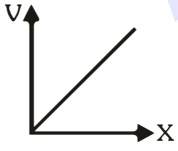
(P) Acceleration $\propto$ v

(B) 

(Q) Acceleration $\propto$ t

(C) 

(R) Acceleration is uniform

(D) 

(S) Acceleration is non uniform

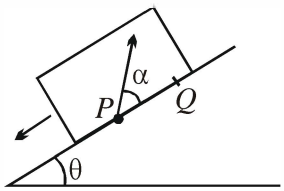
Subjective Type **3 Q. [4 M (0)]**

28. A driver takes 0.20 s to apply the brakes after he sees a need for it. This is called the reaction time of the driver. If he is driving a car at a speed of 54 km/h and the brakes cause a deceleration of 6.0m/s^2, find the distance travelled by the car after he sees the need to put the brakes on

29. A glass wind-screen of adjustable inclination is mounted on a car. The car moves horizontally with a speed of 6 m/s. At what angle α with the vertical should the wind screen be adjusted so that the rain drops falling vertically with 2 m/s strike the wind screen perpendicularly?

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30. A large heavy box is sliding without friction down a smooth plane of inclination θ. From a point *P* on the bottom of the box, a particle is projected inside the box. The initial speed of the particle with respect to box is *u* and the direction of projection makes an angle α with the bottom as shown in figure.



(i) Find the distance along the bottom of the box between the point of projection *P* and the point *Q* where the particle lands. (Assume that the particle does not hit any other surface of the box. Neglect air resistance).

(ii) If the horizontal displacement of the particle as seen by an observer on the ground is zero, find the speed of the box with respect to the ground at the instant when the particle was projected.

GR # UNIT AND DIMENSIONS, VECTORS, KINEMATICS — ANSWER KEY

SECTION-I

Single Correct Answer Type — **6 Q. [3 M (–1)]**

1. (C) **2. (A)** **3. (C)** **4. (A)** **5. (C)**

Multiple Correct Answer Type — **4 Q. [4 M (–1)]**

6. (A,C) **7. (A,B,C,D)** **8. (B, C)** **9. (A,B,D)** **10. (ABC)**

Linked Comprehension Type — (2 Para × 3Q.) (1 Para × 2Q.)[3 M (-1)]

(Single Correct Answer Type)

11. **(B)** **12. (C)** **13. (A)** **14. (B)** **15. (A)**

16. (A) **17. (C)** **18. (D)**

SECTION-II

Numerical Answer Type Question — **3 Q.[3(0)]**

(upto second decimal place)

19. 100.00 **20. 105** **21. 82**

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) — **4 Q. [4 M (0)]**

22. 8 **23. 5** **24. 6** **25. 5**

SECTION-IV

Matrix Match Type (4 × 5) — **2 Q. [8 M (for each entry +2(0)]**

26. (A) P,S,T; (B) P,Q,R,T; (C) P,S,T; (D) P,Q,R,T **27. (A) R; (B) R; (C) Q,S; (D) P,S**

Subjective Type — **3 Q. [4 M (0)]**

28. 21.75 metre. **29. $\tan^{-1}(1/3)$** **30. (i)** $\frac{u^2 \sin 2\alpha}{g\cos\theta}$; **(ii)** $v = \frac{u\cos(\alpha+\theta)}{\cos\theta}$

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